



Contract Playbook Comparison System

Stage 1 — Synthetic-Data Proof of Concept

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Why This Project

Manual contract review doesn't scale — and naive LLM use isn't safe



Slow, non-scaling review

Reviewers read 20–100+ page contracts manually, clause by clause.



Inconsistent outcomes

Different reviewers catch different deviations from the playbook.



No audit trail

Verdicts aren't linked back to the exact source text that produced them.



Hallucination risk

Feeding a whole contract to an LLM risks fabricated, unverifiable answers.



Our approach: a retrieval-augmented pipeline that grounds every verdict in an exact, cited span of the contract — never asking the LLM to judge from memory alone.

Validated market: Ironclad · LinkSquares · Lawgeex · Evisort (Contract Lifecycle Management)

How It's Built

Lightweight, CPU-friendly libraries behind a modular codebase



Tech Stack

- **Embeddings:** sentence-transformers — all-MiniLM-L6-v2
- **Vector index:** FAISS (CPU, IndexFlatIP)
- **Parsing:** pdfplumber (PDF + plain text)
- **Comparison:** LLM API, with rule-based fallback
- **Reporting:** fpdf2 (PDF) + CSV export
- **Demo UI:** Streamlit



Module Layout

ingestion/

PDF/text parsing, chunking, FAISS retrieval

comparison/

Rule-based + LLM comparator, evaluation

reporting/

CSV/PDF export, Streamlit app

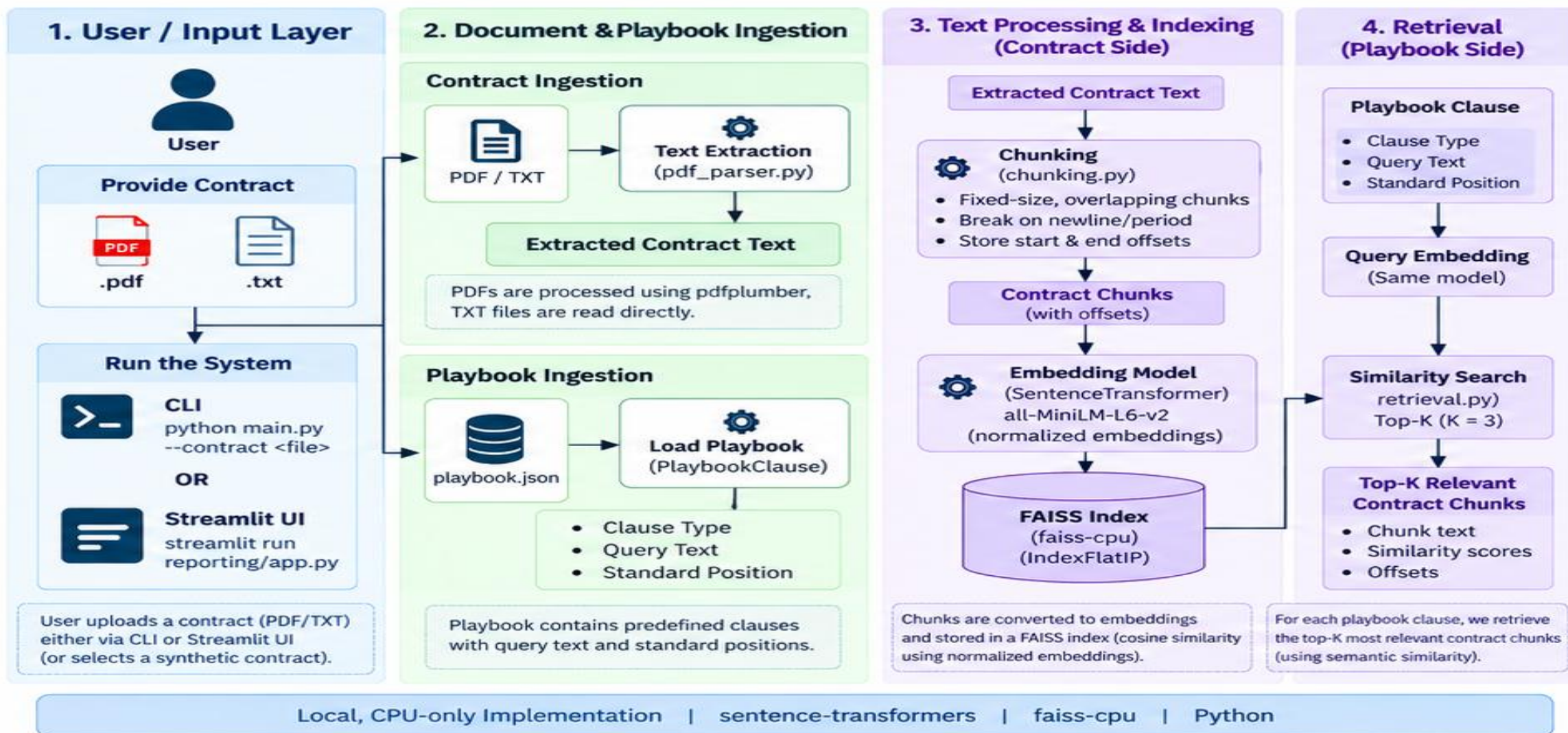
common/

Shared schemas & config — the contract every module relies on

Contract Playbook Comparison – Prototype Implementation

Part 1: Input Processing, Text Representation and Retrieval

From contract and playbook inputs to relevant contract chunks using semantic retrieval

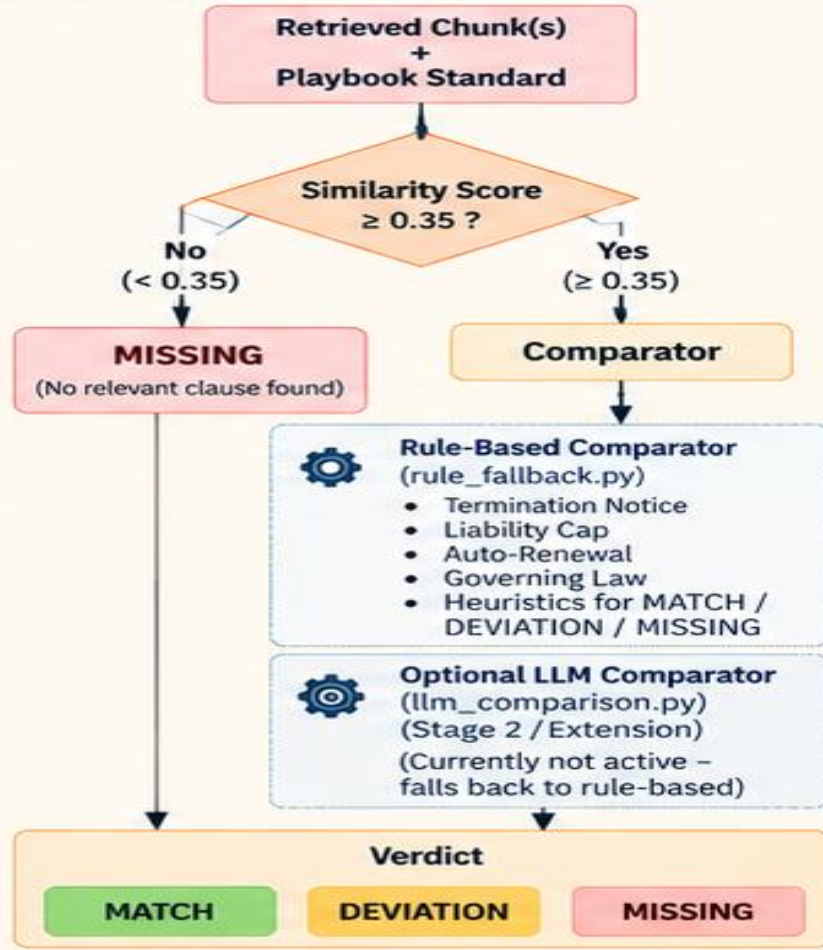


Contract Playbook Comparison – Prototype Implementation

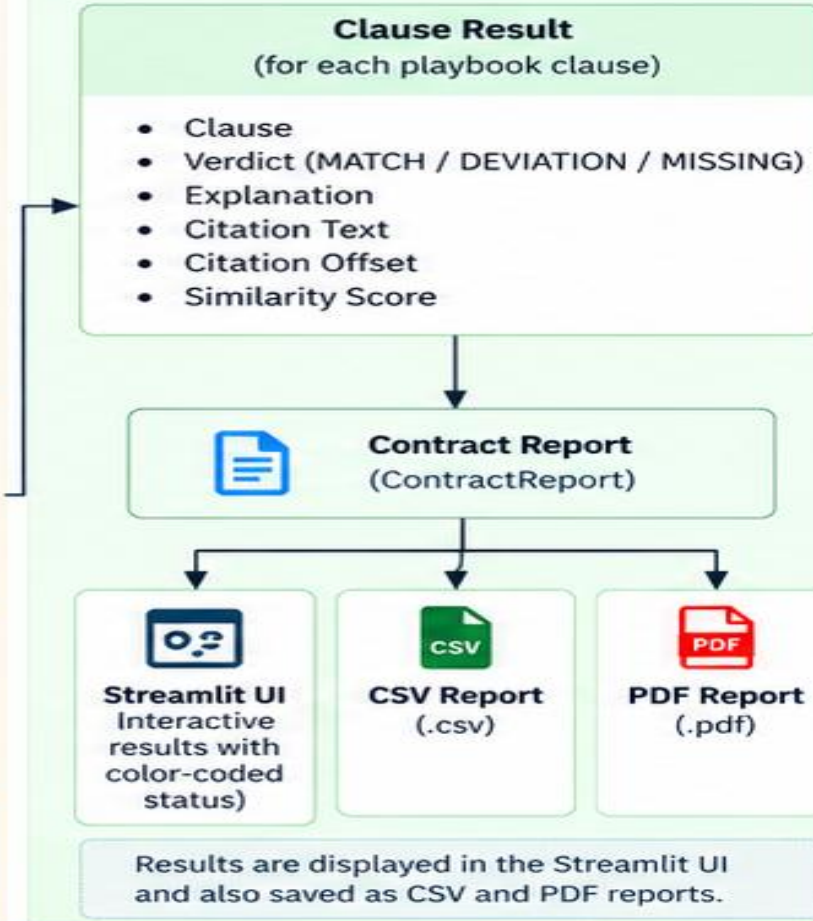
Part 2: Comparison, Results and Evaluation

From retrieved chunks to clause-level decisions, reports and evaluation

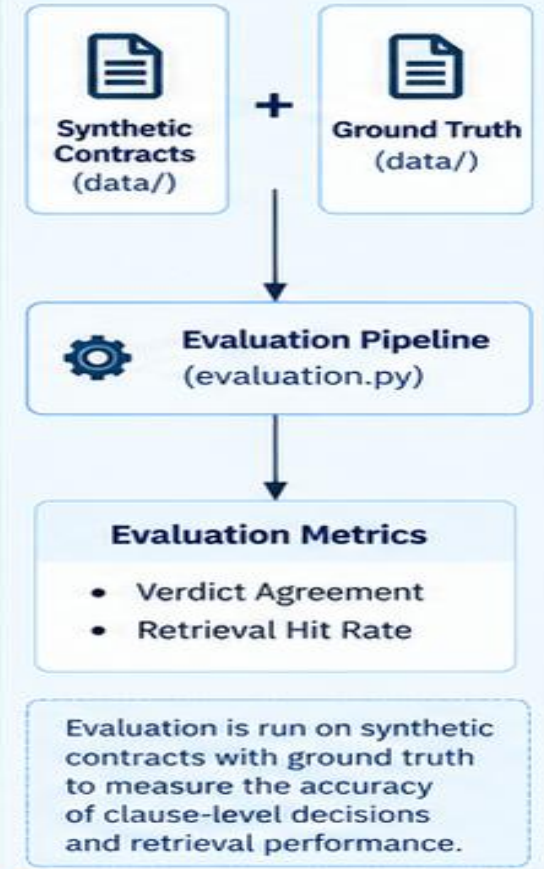
5. Comparison & Decision Logic



6. Output / Results



7. Evaluation (Separate Pipeline)



Generate actionable insights from contract vs. playbook comparison

What's Next

Known limitations today, and the path to Stage 2



Known Limitations

- Comparator uses only the rank-1 candidate — fix identified, not yet applied
- Evaluated on synthetic data only — CUAD not yet integrated
- No hybrid BM25 retrieval yet — semantic-only for now



Stage 2 Roadmap

- Swap in CUAD via the existing pluggable data-source interface
- Add hybrid BM25 + semantic retrieval, with an ablation study
- Scale playbook from 4 to 5–8 clause categories
- Full Recall@k, MRR, and faithfulness evaluation per the proposal

Thank you